

Sensors & Actuators

Robotics

Why Sensors Matter

A robot without sensors is just a machine executing a fixed sequence blindly — it can't react to anything unexpected. Sensors are what let a robot perceive its environment and its own state, closing the loop between 'what I planned to do' and 'what's actually happening.' Choosing the right sensor for a job is one of the most consequential decisions in a robot's design, because it directly limits what the robot can ever know about the world.

Common Sensor Types

Sensors used in robotics generally fall into a few families, each answering a different question:

- Proximity/distance sensors (ultrasonic, infrared, LiDAR) — 'how far is the nearest object?'
Used for obstacle avoidance and mapping.
- Touch/contact sensors (limit switches, bump sensors, force-sensitive resistors) — 'did I just touch or collide with something?'
- Light sensors (photoresistors, IR reflectance sensors) — used for line-following and detecting light/dark surfaces.
- Inertial sensors (accelerometers, gyroscopes, combined in an IMU) — 'how am I oriented, and how am I accelerating or rotating?' Critical for balance and navigation.
- Encoders — measure how far a motor shaft or wheel has rotated, letting the robot estimate distance traveled (odometry).
- Cameras / vision sensors — capture images for more complex perception like object recognition or line detection with more detail than a simple light sensor.

Motors and Actuators

Actuators are the components that convert electrical energy into physical motion or force — they're the robot's muscles. The right choice depends on the type of motion needed and how precisely it must be controlled:

- DC motors — simple, continuous rotation, speed roughly proportional to voltage; common for driving wheels.
- Servo motors — DC motors with built-in position feedback and control circuitry, allowing precise angle control (commonly 0-180°); used for joints and steering.
- Stepper motors — move in fixed, precise increments ('steps') per electrical pulse, giving accurate position control without needing a feedback sensor; common in 3D printers and camera gimbals.

- Solenoids — simple electromagnetic actuators that produce linear push/pull motion, often used for grippers or latches.
- Motor drivers/H-bridges — the electronic circuit between a low-power microcontroller and a higher-power motor, needed because microcontroller pins can't supply enough current to run a motor directly.

Reading Sensor Data

Raw sensor readings are rarely usable straight from the pin — real-world sensors are noisy, and a robot needs to turn a voltage or digital signal into a meaningful number before it can make decisions from it.

- Analog vs. digital signals: many sensors (like a simple IR reflectance sensor) output an analog voltage that must be read through an analog-to-digital converter (ADC); others output clean digital HIGH/LOW signals.
- Calibration: taking a few known reference readings (e.g. sensor pointed at black vs. white surface) to translate raw values into meaningful units.
- Filtering/smoothing: averaging several readings over time (or using techniques like a moving average) to reduce noise before acting on a value.
- Sampling rate: how often you read a sensor matters — reading too infrequently can miss fast-changing events like a sudden collision.

Putting It Together: A Simple Sense-Act Example

Consider a robot with one ultrasonic distance sensor mounted on the front and two DC motors driving its wheels. The control loop reads the distance sensor, and if the reading falls below a threshold (say, 15 cm), it stops the forward motors and reverses briefly before turning. This simple pattern — read sensor, compare to a threshold, decide an action, drive an actuator — is the building block for almost every more sophisticated robot behavior covered later in this course, including full autonomous navigation.

Learn More

SparkFun — Sensors Overview

<https://www.sparkfun.com/>

Arduino Official Documentation

<https://docs.arduino.cc/>

NPTEL — Sensors and Actuators

<https://nptel.ac.in/>